

1

i

ii

2

i

```
data = read.csv("data/slumptest.csv", sep=",")
head(data)
```

	Water	CompressiveStrength
1	210	34.99
2	180	41.14
3	179	41.81
4	179	42.08
5	220	26.82
6	202	25.21

```
model = lm(CompressiveStrength ~ Water, data=data)
print(paste("Beta0 =" , model$coefficients[1]))
```

```
[1] "Beta0 = 55.4824064174304"
```

```
print(paste("Beta1 =" , model$coefficients[2]))
```

```
[1] "Beta1 = -0.0986112998623879"
```

ii

If there is no water content then the compressive strength of concrete is on average 55.482.

On average, the compressive strength of the concrete decreases by 0.09861 per unit increase of water content

iii

```
vcov(model)
```

	(Intercept)	Water
(Intercept)	54.7279850	-0.274712638
Water	-0.2747126	0.001393292

iv

```
confint(model, level=0.95)
```

	2.5 %	97.5 %
(Intercept)	40.8070888	70.1577241
Water	-0.1726577	-0.0245649

Thus we are 95% confident that when the water content is 0 then the compressive strength of the concrete is between [40.807, 70.1577]

Thus we are 95% confident that the compressive strength of the concrete increases on average $[-0.1726577, -0.0245649]$ per unit increase of water content.

v

vi

```
predict(model, data.frame(Water=230))
```

	1
	32.80181

vii

```
predict(model, data.frame(Water=230), interval="predict", level=0.95)
```

	fit	lwr	upr
1	32.80181	17.42295	48.18067